20

Structure Fixed Controllers

In this chapter we focus on the problem of designing optimal controllers with controller structures restricted; for instance, the controller may be limited to be a state feedback or a constant output feedback or a fixed order dynamic controller. We shall be interested in deriving some explicit necessary conditions that an optimal fixed structure controller ought to satisfy. The fundamental idea is to formulate our optimal control problems as some constrained minimization problems. Then the first-order Lagrange multiplier necessary conditions for optimality are applied to derive our optimal controller formulae. Readers should keep in mind that our purpose here is to introduce the method, not to try to solve as many problems as possible. Hence, we will try to be concise but clear. In section 20.1, we will review some Lagrange multiplier optimization methods. Then these tools will be used in section 20.2 to solve a fixed order $\mathcal{H}_2$ optimal controller problem.

20.1 Lagrange Multiplier Method

In this section, we consider the constrained minimization problem. The results quoted below are standard and can be found in any references given at the end of the chapter.

Let $f(x) := f(x_1, x_2, \ldots, x_n) \in \mathbb{R}$ be a real valued function defined on a set $S \subset \mathbb{R}^n$. A point $x_0 \in \mathbb{R}^n$ in S is said to be a (global) *minimum point* of f on S if

$$f(x) \geq f(x_0)$$

for all points $x \in S$. A point $x_0 \in S$ is said to be a *local minimum point* of f on S if there is a neighborhood N of x_0 such that $f(x) \geq f(x_0)$ for all points $x \in N$.

We will be particularly interested in the case where the set S is described by a set of functions, $h_i(x) = 0, i = 1, 2, \ldots, m$ and $m < n$ or equivalently

$$H(x) := \left[\begin{array}{cccc} h_1(x) & h_2(x) & \ldots & h_m(x) \end{array} \right] = 0.$$

Hence we will focus on the local necessary (and sufficient) conditions for the following problem:

$$\begin{cases} \text{minimize } f(x) \\ \text{subject to } H(x) = 0. \end{cases} \tag{20.1}$$

We shall assume that $f(x)$ and $h_i(x)$ are differentiable and denote

$$\nabla f(x) := \left[\begin{array}{c} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{array} \right]$$

$$\nabla H(x) := \left[\begin{array}{cccc} \nabla h_1(x) & \nabla h_2(x) & \cdots & \nabla h_m(x) \end{array} \right] = \left[\begin{array}{cccc} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_2}{\partial x_1} & \cdots & \frac{\partial h_m}{\partial x_1} \\ \frac{\partial h_1}{\partial x_2} & \frac{\partial h_2}{\partial x_2} & \cdots & \frac{\partial h_m}{\partial x_2} \\ \vdots & \vdots & & \vdots \\ \frac{\partial h_1}{\partial x_n} & \frac{\partial h_2}{\partial x_n} & \cdots & \frac{\partial h_m}{\partial x_n} \end{array} \right].$$

Definition 20.1 A point $x_0 \in \mathbb{R}^n$ satisfying the constraints $H(x_0) = 0$ is said to be a *regular point* of the constraints if $\nabla H(x_0)$ has full column rank (m); equivalently, let $\phi(x) := H(x)z$ for $z \in \mathbb{R}^m$, and then $\nabla \phi(x_0) = \nabla H(x_0)z = 0$ has the unique solution $z = 0$.

Theorem 20.1 *Suppose that $x_0 \in \mathbb{R}^n$ is a local minimum of the $f(x)$ subject to the constraints $H(x) = 0$ and suppose further that x_0 is a regular point of the constraints. Then there exists a unique multiplier*

$$\Lambda = \left[\begin{array}{c} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_m \end{array} \right] \in \mathbb{R}^m$$

such that, if we set $F(x) = f(x) + H(x)\Lambda$, then $\nabla F(x_0) = 0$, i.e.,

$$\nabla F(x_0) = \nabla f(x_0) + \nabla H(x_0)\Lambda = 0.$$

In the case where the regular point conditions are either not satisfied or hard to verify, we have the following alternative.

Theorem 20.2 *Suppose that $x_0 \in \mathbb{R}^n$ is a local minimum of $f(x)$ subject to the constraints $H(x) = 0$. Then there exists*

$$\begin{bmatrix} \lambda_0 \\ \Lambda \end{bmatrix} \in \mathbb{R}^{m+1}$$

such that $\lambda_0 \nabla f(x_0) + \nabla H(x_0)\Lambda = 0$.

Remark 20.1 Although some second order necessary and sufficient conditions for local minimality can be given, they are usually not very useful in our applications for the reason that they are very hard to verify except in some very special cases. Furthermore, even if some sufficient conditions can be verified, it is still not clear whether the minima is global. It is a common practice in many applications that the first-order necessary conditions are used to derive some necessary conditions for the existence of a minima. Then find a solution (or solutions) from these necessary conditions and check if the solution(s) satisfies our objectives regardless of the solution(s) being a global minima or not. ♡

In most of the control applications, constraints are given by a symmetric matrix function, and in this case, we have the following lemma.

Lemma 20.3 *Let $T(x) = T(x)^* \in \mathbb{R}^{l \times l}$ be a symmetric matrix function and let $x_0 \in \mathbb{R}^n$ be such that $T(x_0) = 0$. Then x_0 is a regular point of the constraints $T(x) = T(x)^* = 0$ if, for $P = P^* \in \mathbb{R}^{l \times l}$, $\nabla\,\mathrm{Trace}(T(x_0)P) = 0$ has the unique solution $P = 0$.*

Proof. Since $T(x) = [t_{ij}(x)]$ is a symmetric matrix, $t_{ij} = t_{ji}$ and the effective constraints for $T(x) = 0$ are given by the $l(l+1)/2$ equations, $t_{ij}(x) = 0$ for $i = 1, 2, \ldots, l$ and $i \leq j \leq l$. By definition, x_0 is a regular point for the effective constraints ($t_{ij}(x) = 0$ for $i = 1, 2, \ldots, l$ and $i \leq j \leq l$) if the following equation has the unique solution $p_{ij} = 0$ for $i = 1, 2, \ldots, l$ and $i \leq j \leq l$:

$$\psi(x_0) := \sum_{i=1}^{l} \nabla t_{ii} p_{ii} + \sum_{i=1}^{l} \sum_{j=i+1}^{l} 2\nabla t_{ij} p_{ji} = 0.$$

Now the result follows by defining $p_{ij} := p_{ji}$ and by noting that $\psi(x_0)$ can be written as

$$\psi(x_0) = \nabla \left(\sum_{i=1}^{l} \sum_{j=1}^{l} t_{ij} p_{ji} \right) = \nabla\,\mathrm{Trace}(T(x_0)P)$$

with $P = P^*$. □

Corollary 20.4 *Suppose that $x_0 \in \mathbb{R}^n$ is a local minimum of $f(x)$ subject to the constraints $T(x) = 0$ where $T(x) = T(x)^* \in \mathbb{R}^{l \times l}$ and suppose further that x_0 is a regular* *point of the constraints. Then there exists a unique multiplier* $P = P^* \in \mathbb{R}^{l\times l}$ *such that if we set* $F(x) = f(x) + \mathrm{Trace}(T(x)P)$*, then* $\nabla F(x_0) = 0$*, i.e.,*

$$\nabla F(x_0) = \nabla f(x_0) + \nabla\,\mathrm{Trace}(T(x_0)P) = 0.$$

In general, in the case where a local minimal point x_0 is not necessarily a regular point, we have the following corollary.

Corollary 20.5 *Suppose that* $x_0 \in \mathbb{R}^n$ *is a local minimum of* $f(x)$ *subject to the constraints* $T(x) = 0$ *where* $T(x) = T(x)^* \in \mathbb{R}^{l\times l}$*. Then there exist* $0 \neq (\lambda_0, P) \in \mathbb{R}\times\mathbb{R}^{l\times l}$ *with* $P = P^*$ *such that*

$$\lambda_0 \nabla f(x_0) + \nabla\,\mathrm{Trace}(T(x_0)P) = 0.$$

Remark 20.2 We shall also note that the variable $x \in \mathbb{R}^n$ may be more conveniently given in terms of a matrix $X \in \mathbb{R}^{k\times q}$, i.e., we have

$$x = \mathrm{Vec}X := \begin{bmatrix} x_{11} \\ \vdots \\ x_{k1} \\ x_{12} \\ \vdots \\ x_{k2} \\ \vdots \\ x_{kq} \end{bmatrix}.$$

Then

$$\nabla F(x) := \begin{bmatrix} \frac{\partial F(x)}{\partial x_{11}} \\ \vdots \\ \frac{\partial F(x)}{\partial x_{kq}} \end{bmatrix} = 0$$

is equivalent to

$$\frac{\partial F(x)}{\partial X} := \begin{bmatrix} \frac{\partial F(x)}{\partial x_{11}} & \frac{\partial F(x)}{\partial x_{12}} & \cdots & \frac{\partial F(x)}{\partial x_{1q}} \\ \frac{\partial F(x)}{\partial x_{21}} & \frac{\partial F(x)}{\partial x_{22}} & \cdots & \frac{\partial F(x)}{\partial x_{2q}} \\ \vdots & \vdots & & \vdots \\ \frac{\partial F(x)}{\partial x_{k1}} & \frac{\partial F(x)}{\partial x_{k2}} & \cdots & \frac{\partial F(x)}{\partial x_{kq}} \end{bmatrix} = 0.$$

This later expression will be used throughout in the sequel. ♡

As an example, let us consider the following $\mathcal{H}_2$ norm minimization with constant state feedback: the dynamic system is given by

$$\begin{aligned} \dot{x} &= Ax + B_1 w + B_2 u \\ z &= C_1 x + D_{12} u, \end{aligned}$$

and the feedback $u = Fx$ is chosen so that $A + B_2F$ is stable and

$$J_0 = \|T_{zw}\|_2^2$$

is minimized. For simplicity, we shall assume that $D_{12}^*D_{12} = I$ and $D_{12}^*C_1 = 0$. It is routine to verify that $J_0 = \text{Trace}(B_1B_1^*X)$ where $X = X^* \geq 0$ satisfies

$$T(X,F) := X(A + B_2F) + (A + B_2F)^*X + (C_1 + D_{12}F)^*(C_1 + D_{12}F) = 0.$$

Hence the optimal control problem becomes a constrained minimization problem, and we can use the Lagrange multipliers method outlined above. (Note that in this case, the variable x takes the form $x = \begin{bmatrix} \text{Vec}X \\ \text{Vec}F \end{bmatrix}$). Let

$$J(X,F) := J_0 + \text{Trace}(T(X,F)P)$$

with $P = P^*$. We first verify the regularity conditions: the equation

$$\nabla\, \text{Trace}(T(X,F)P) = 0$$

or, equivalently,

$$\begin{bmatrix} \frac{\partial\, \text{Trace}(T(X,F)P)}{\partial X} \\ \frac{\partial\, \text{Trace}(T(X,F)P)}{\partial F} \end{bmatrix} = \begin{bmatrix} P(A + B_2F)^* + (A + B_2F)P \\ 2(B_2^*X + F)P \end{bmatrix} = 0$$

has a unique solution $P = 0$ since $A + B_2F$ is assumed to be stable at the minimum point. Hence regularity conditions are satisfied. Now the necessary condition for local optimum can be applied:

$$\frac{\partial J(X,F)}{\partial X} = B_1B_1^* + P(A + B_2F)^* + (A + B_2F)P = 0 \tag{20.2}$$

$$\frac{\partial J(X,F)}{\partial F} = 2(B_2^*X + F)P = 0 \tag{20.3}$$

$$\frac{\partial J(X,F)}{\partial P} = X(A + B_2F) + (A + B_2F)^*X + (C_1 + D_{12}F)^*(C_1 + D_{12}F) = 0. \tag{20.4}$$

It should be pointed out that, in general, we cannot conclude from equation (20.3) that $B_2^*X + F = 0$. Care must be exercised to arrive at such a conclusion. For example, if we assume that B_1 is square and nonsingular, then we know that if F is such that $A + B_2F$ is stable, then $P > 0$. Hence we have

$$F = -B_2^*X,$$

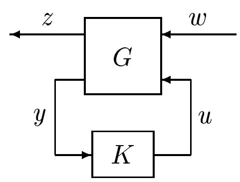
and we substitute this relation into equation (20.4) and get the familiar Riccati equation:

$$XA + A^*X - XB_2B_2^*X + C_1^*C_1 = 0.$$

This Riccati equation has a stabilizing solution if (A, B_2) is stabilizable and if (C_1, A) has no unobservable modes on the imaginary axis. Indeed, in this case, the controller thus obtained is a global optimal control law.

20.2 Fixed Order Controllers

In this section, we shall use the Lagrange multiplier method to derive some necessary conditions for a fixed-order controller that minimizes an $\mathcal{H}_2$ performance. We shall again consider a standard system setup



where the system G is n-th order with a realization given by

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{array}\right].$$

For simplicity, we shall assume that

(i) (A, B_1) is stabilizable and (C_1, A) is detectable;

(ii) (A, B_2) is stabilizable and (C_2, A) is detectable;

(iii) $D_{12}^* \begin{bmatrix} C_1 & D_{12} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}$;

(iv) $\begin{bmatrix} B_1 \\ D_{21} \end{bmatrix} D_{21}^* = \begin{bmatrix} 0 \\ I \end{bmatrix}$.

We shall be interested in the following fixed order $\mathcal{H}_2$ optimal controller problem: *given an integer $n_c \leq n$, find an n_c-th order controller*

$$K = \left[\begin{array}{c|c} A_c & B_c \\ \hline C_c & 0 \end{array}\right]$$

that internally stabilizes the system G and minimizes the $\mathcal{H}_2$ norm of the transfer matrix T_{zw}. For technical reasons, we will further assume that the realization of the controller is minimal, i.e., *(A_c, B_c) is controllable and (C_c, A_c) is observable.*

Suppose a such controller exists, and then the closed loop transfer matrix T_{zw} can be written as

$$T_{zw} = \left[\begin{array}{cc|c} A & B_2C_c & B_1 \\ B_cC_2 & A_c & B_cD_{21} \\ \hline C_1 & D_{12}C_c & 0 \end{array}\right] =: \left[\begin{array}{c|c} \tilde{A} & \tilde{B} \\ \hline \tilde{C} & 0 \end{array}\right]$$

with $\tilde{A}$ stable. Moreover,

$$\|T_{zw}\|_2^2 = \mathrm{Trace}(\tilde{B}\tilde{B}^*\tilde{X}) \tag{20.5}$$

where $\tilde{X}$ is the observability Gramian of T_{zw}:

$$\tilde{X}\tilde{A} + \tilde{A}^*\tilde{X} + \tilde{C}^*\tilde{C} = 0. \tag{20.6}$$

Theorem 20.6 *Suppose (A_c, B_c, C_c) is a controllable and observable triple and $K = \left[\begin{array}{c|c} A_c & B_c \\ \hline C_c & 0 \end{array}\right]$ internally stabilizes the system G and minimizes the norm T_{zw}. Then there exist $n \times n$ nonnegative definite matrices X, Y, $\hat{X}$, and $\hat{Y}$ such that A_c, B_c, and C_c are given by*

$$\begin{aligned} A_c &= \Gamma(A - B_2B_2^*X - YC_2^*C_2)\Pi^* && (20.7)\\ B_c &= \Gamma Y C_2^* && (20.8)\\ C_c &= -B_2^*X\Pi^* && (20.9) \end{aligned}$$

for some factorization

$$\hat{Y}\hat{X} = \Pi^*M\Gamma, \quad \Gamma\Pi^* = I_{n_c}$$

with M positive-semisimple[1] *and such that with $\tau := \Pi^*\Gamma$ and $\tau_\perp := I_n - \tau$ the following conditions are satisfied:*

$$\begin{aligned} 0 &= A^*X + XA - XB_2B_2^*X + \tau_\perp^*XB_2B_2^*X\tau_\perp + C_1^*C_1 && (20.10)\\ 0 &= AY + YA^* - YC_2^*C_2Y + \tau_\perp YC_2^*C_2Y\tau_\perp^* + B_1B_1^* && (20.11)\\ 0 &= (A - YC_2^*C_2)^*\hat{X} + \hat{X}(A - YC_2^*C_2) + XB_2B_2^*X - \tau_\perp^*XB_2B_2^*X\tau_\perp && (20.12)\\ 0 &= (A - B_2B_2^*X)\hat{Y} + \hat{Y}(A - B_2B_2^*X)^* + YC_2^*C_2Y - \tau_\perp YC_2^*C_2Y\tau_\perp^* && (20.13) \end{aligned}$$

$$\mathrm{rank}\ \hat{X} = \mathrm{rank}\ \hat{Y} = \mathrm{rank}\ \hat{Y}\hat{X} = n_c.$$

Proof. The problem can be viewed as a constrained minimization problem with the objective function given by equation (20.5) and with constraints given by equation (20.6). Let $\tilde{Y} = \tilde{Y}^* \in \mathbb{R}^{(n+n_c)\times(n+n_c)}$ and denote

$$J_1 = \mathrm{Trace}\left\{(\tilde{X}\tilde{A} + \tilde{A}^*\tilde{X} + \tilde{C}^*\tilde{C})\tilde{Y}\right\}.$$

Then

$$\frac{\partial J_1}{\partial \tilde{X}} = \tilde{Y}\tilde{A}^* + \tilde{A}\tilde{Y} = 0$$

has the unique solution $\tilde{Y} = 0$ since $\tilde{A}$ is assumed to be stable. Hence the regularity conditions are satisfied and the Lagrange multiplier method can be applied. Form the Lagrange function J as

$$J := \mathrm{Trace}(\tilde{B}\tilde{B}^*\tilde{X}) + J_1$$

[1] A matrix M is called *positive semisimple* if it is similar to a positive definite matrix.

and partition $\tilde{X}$ and $\tilde{Y}$ as

$$\tilde{X} = \begin{bmatrix} X_{11} & X_{12} \\ X_{12}^* & X_{22} \end{bmatrix}, \quad \tilde{Y} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{12}^* & Y_{22} \end{bmatrix}.$$

The necessary conditions for (A_c, B_c, C_c) to be a local minima are

$$\begin{aligned} \frac{\partial J}{\partial A_c} &= 2(X_{12}^* Y_{12} + X_{22} Y_{22}) = 0 && (20.14) \\ \frac{\partial J}{\partial B_c} &= 2(X_{22} B_c + X_{12}^* Y_{11} C_2^* + X_{22} Y_{12}^* C_2^*) = 0 && (20.15) \\ \frac{\partial J}{\partial C_c} &= 2(B_2^* X_{11} Y_{12} + B_2^* X_{12} Y_{22} + C_c Y_{22}) = 0 && (20.16) \\ \frac{\partial J}{\partial \tilde{Y}} &= \tilde{X}\tilde{A} + \tilde{A}^*\tilde{X} + \tilde{C}^*\tilde{C} = 0 && (20.17) \\ \frac{\partial J}{\partial \tilde{X}} &= \tilde{Y}\tilde{A}^* + \tilde{A}\tilde{Y} + \tilde{B}^*\tilde{B} = 0. && (20.18) \end{aligned}$$

It is clear that $\tilde{X} \geq 0$ and $\tilde{Y} \geq 0$ since $\tilde{A}$ is stable. Equations (20.17) and (20.18) can be written as

$$\begin{aligned} 0 &= X_{11}A + A^*X_{11} + X_{12}B_cC_2 + C_2^*B_c^*X_{12}^* + C_1^*C_1 && (20.19) \\ 0 &= X_{12}A_c + A^*X_{12} + X_{11}B_2C_c + C_2^*B_c^*X_{22} && (20.20) \\ 0 &= X_{22}A_c + A_c^*X_{22} + X_{12}^*B_2C_c + C_c^*B_2^*X_{12} + C_c^*C_c && (20.21) \\ 0 &= AY_{11} + Y_{11}A^* + B_2C_cY_{12}^* + Y_{12}C_c^*B_2^* + B_1B_1^* && (20.22) \\ 0 &= AY_{12} + Y_{12}A_c^* + B_2C_cY_{22} + Y_{11}C_2^*B_c^* && (20.23) \\ 0 &= A_cY_{22} + Y_{22}A_c^* + B_cC_2Y_{12} + Y_{12}^*C_2^*B_c^* + B_cB_c^*. && (20.24) \end{aligned}$$

For clarity, we shall present the proof in three steps:

1. $X_{22} > 0$ and $Y_{22} > 0$: We show first that $X_{22} > 0$. Since $\tilde{X} \geq 0$, we have $X_{22} \geq 0$ and $X_{22}X_{22}^+X_{12}^* = X_{12}^*$ by Lemma 2.16. Hence equation (20.21) can be written as

$$0 = X_{22}(A_c + X_{22}^+X_{12}^*B_2C_c) + (A_c + X_{22}^+X_{12}^*B_2C_c)^*X_{22} + C_c^*C_c.$$

   Since (C_c, A_c) is observable by assumption, $(C_c, A_c + X_{22}^+X_{12}^*B_2C_c)$ is also observable. Now it follows from the Lyapunov theorem (Lemma 3.18) that $X_{22} > 0$. $Y_{22} > 0$ follows by a similar argument.

2. formula for A_c, B_c, C_c, Γ, and Π: given $X_{22} > 0$ and $Y_{22} > 0$, we define

$$\begin{aligned} \Gamma &:= -X_{22}^{-1}X_{12}^* && (20.25) \\ \Pi &:= Y_{22}^{-1}Y_{12}^* && (20.26) \\ \tau &:= \Pi^*\Gamma && (20.27) \end{aligned}$$

$$
\begin{aligned}
\hat{X} &:= X_{12}X_{22}^{-1}X_{12}^* \geq 0 && (20.28)\\
\hat{Y} &:= Y_{12}Y_{22}^{-1}Y_{12}^* \geq 0 && (20.29)\\
X &:= X_{11} - X_{12}X_{22}^{-1}X_{12}^* && (20.30)\\
Y &:= Y_{11} - Y_{12}Y_{22}^{-1}Y_{12}^*. && (20.31)
\end{aligned}
$$

Then it follows from equation (20.14) that

$$\Gamma\Pi^* = I$$

and $\tau^2 = \Pi^*\Gamma\Pi^*\Gamma = \Pi^*\Gamma = \tau$. It also follows from $\tilde{X} \geq 0$ and $\tilde{Y} \geq 0$ that $X \geq 0$ and $Y \geq 0$. Moreover, from equation (20.14), we have

$$n_c = \text{rank } X_{22} \leq \text{rank } X_{12} \leq n_c.$$

This implies that rank $X_{12} = n_c =$ rank Y_{12} and

$$\text{rank } \hat{X} = \text{rank } X_{12} = n_c = \text{rank } \hat{Y} = \text{rank } \hat{Y}\hat{X}.$$

The product of $\hat{Y}\hat{X}$ can be factored as

$$\hat{Y}\hat{X} = \Pi^*(-Y_{12}^*X_{12})\Gamma = \Pi^*Y_{22}X_{22}\Gamma,$$

and

$$M := Y_{22}X_{22} = Y_{22}^{1/2}(Y_{22}^{1/2}X_{22}Y_{22}^{1/2})Y_{22}^{-1/2}$$

is a positive semisimple matrix.

Using these formulae in equations (20.15) and (20.16), we get

$$
\begin{aligned}
B_c &= -(X_{22}^{-1}X_{12}^*Y_{11} + Y_{12}^*)C_2^* = (\Gamma Y_{11} - (\Gamma\Pi^*)Y_{12}^*)C_2^*\\
&= \Gamma(Y_{11} - \Pi^*Y_{12}^*)C_2^*\\
&= \Gamma Y C_2^*
\end{aligned}
$$

and

$$
\begin{aligned}
C_c &= -B_2^*(X_{11}Y_{12}Y_{22}^{-1} + X_{12}) = -B_2^*(X_{11} + X_{12}\Gamma)\Pi^*\\
&= -B_2^*X\Pi^*.
\end{aligned}
$$

The formula for A_c follows from (20.24)-$\Gamma\times$(20.23) and some algebra.

3. Equations for X, Y, $\hat{X}$, and $\hat{Y}$: Equations (20.12) and (20.13) follow by substituting A_c, B_c, and C_c into (20.20)$\times\Gamma$ and (20.23)$\times\Pi$ and by using the fact that $\hat{X} = \tau\hat{X}$ and $\hat{Y} = \tau\hat{Y}$. Finally, equations (20.10) and (20.11) follow from equations (20.19) and (20.22) with some tedious algebra.

□

Remark 20.3 It is interesting to note that if the full order controller $n_c = n$ is considered, then we have $\tau = I$ and $\tau_\perp = 0$. In that case, equations (20.10) and (20.11) become standard $\mathcal{H}_2$ Riccati equations:

$$\begin{aligned} 0 &= A^*X + XA - XB_2B_2^*X + C_1^*C_1 \\ 0 &= AY + YA^* - YC_2^*C_2Y + B_1B_1^* \end{aligned}$$

Moreover, there exist unique stabilizing solutions $X \geq 0$ and $Y \geq 0$ to these two Riccati equations such that $A - B_2B_2^*X$ and $A - YC_2^*C_2$ are stable. Using these facts, we get that equations (20.12) and (20.13) have unique solutions:

$$\begin{aligned} \hat{X} &= \int_0^\infty e^{(A-YC_2^*C_2)^*t} XB_2B_2^*X e^{(A-YC_2^*C_2)t} dt \geq 0 \\ \hat{Y} &= \int_0^\infty e^{(A-B_2B_2^*X)t} YC_2^*C_2Y e^{(A-B_2B_2^*X)^*t} dt \geq 0. \end{aligned}$$

It is a fact that $\hat{X}$ is nonsingular iff $(B_2^*X, A - YC_2^*C_2)$ is observable and that $\hat{Y}$ is nonsingular iff $(A - B_2B_2^*X, YC_2^*)$ is controllable, or equivalently iff

$$K_o := \left[\begin{array}{c|c} A - B_2B_2^*X - YC_2^*C_2 & YC_2^* \\ \hline -B_2^*X & 0 \end{array}\right]$$

is controllable and observable. (Note that K_o is known to be the optimal $\mathcal{H}_2$ controller from Chapter 14). Furthermore, if $\hat{X}$ and $\hat{Y}$ are nonsingular, we can indeed find Γ and Π such that $\Gamma\Pi^* = I_n$. In fact, in this case, Γ and Π are both square and $\Gamma = (\Pi^*)^{-1}$. Hence, we have

$$\begin{aligned} K &= \left[\begin{array}{c|c} A_c & B_c \\ \hline C_c & 0 \end{array}\right] = \left[\begin{array}{c|c} \Gamma(A - B_2B_2^*X - YC_2^*C_2)\Gamma^{-1} & \Gamma YC_2^* \\ \hline -B_2^*X\Gamma^{-1} & 0 \end{array}\right] \\ &= \left[\begin{array}{c|c} A - B_2B_2^*X - YC_2^*C_2 & YC_2^* \\ \hline -B_2^*X & 0 \end{array}\right] = K_o \end{aligned}$$

i.e., if $\hat{X}$ and $\hat{Y}$ are nonsingular or, equivalently, if optimal controller K_o is controllable and observable as we assumed, then Theorem 20.6 generates the optimal controller. However, in general, K_o is not necessarily minimal; hence Theorem 20.6 will not be applicable.

It is possible to derive some similar results to Theorem 20.6 without assuming the minimality of the optimal controller by using pseudo-inverse in the derivations, but that, in general, is much more complicated. An alternative solution to this dilemma would be simply by direct testing: if a given n_c does not generate a controllable and observable controller, then lower n_c and try again. ♡

Remark 20.4 We should also note that although we have the necessary conditions for a reduced order optimal controller, it is generally hard to solve these coupled equations although some ad hoc homotopy algorithm might be used to find a local minima. ♡

Remark 20.5 This method can also be used to derive the $\mathcal{H}_\infty$ results presented in the previous chapters. The interested reader should consult the references for details. It should be pointed out that this method suffers a severe deficiency: global results are hard to find. This is due to (a) only first order necessary conditions can be relatively easily derived; (b) the controller order must be fixed; hence even if a fixed-order optimal controller can be found, it may not be optimal over all stabilizing controllers. ♡

20.3 Notes and References

Optimization using the Lagrange multiplier can be found in any standard optimization textbook. In particular, the book by Hestenes [1975] contains the finite dimensional case, and the one by Luenberger [1969] contains both finite and infinite dimensional cases. The Lagrange multiplier method has been used extensively by Hyland and Bernstein [1984], Bernstein and Haddard [1989], and Skelton [1988] in control applications. Theorem 20.6 was originally shown in Hyland and Bernstein [1984].